

Divy Patel

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SUMMARY

Machine Learning Engineer with over 3 years of experience in designing and deploying machine learning models and microservices. Demonstrated expertise in deep learning, autonomous driving technologies, and optimization of ML algorithms for embedded systems. Proven ability to implement state-of-the-art models using frameworks like TensorFlow, PyTorch, and scikit-learn. Adept in cloud technologies, data processing, and improving system performance. Eager to contribute to cutting-edge AI/ML projects in a dynamic team environment.

SKILLS:

- **Programming Languages:** Python (including Python 3.x, Pandas, NumPy, SciPy), C++, Java, JavaScript, SQL
- **Machine Learning Frameworks:** TensorFlow, PyTorch, Keras, scikit-learn, MXNet
- **Deep Learning:** CNNs, RNNs, Foundation Models, LLMs, GANs, Diffusion Models
- **Database Management:** SQL (MySQL, PostgreSQL, MS SQL Server, SQLite, ChromaDB), NoSQL (MongoDB).
- **Cloud Technologies:** AWS (EC2, S3, RDS), Azure, Google Cloud Platform.
- **DevOps Tools:** Docker, Kubernetes, Terraform, Jenkins, Ansible, Git.
- **API Development:** RESTful Services, GraphQL APIs.
- **Agile Methodologies:** Agile, Scrum.

EXPERIENCE:

Machine Learning Engineer- PetSmart, Phoenix, Arizona

May 2023 - Current

- Developed and fine-tuned deep learning models to enhance image recognition and classification for pet products, improving the accuracy of automated product tagging by 20%.
- Integrated and optimized large-scale language models to enhance natural language processing capabilities for pet-related customer queries, improving response accuracy and speed.
- Designed algorithms for analyzing and predicting customer behavior patterns based on purchase history, enabling targeted marketing strategies and increasing customer engagement by 15%.
- Utilized TensorFlow and PyTorch to build and deploy custom models for real-time inventory management and demand forecasting, resulting in a 10% reduction in stockouts.
- Implemented Spark and Hive for processing large datasets from the company's extensive customer and inventory databases, reducing data processing times by 15%.
- Enhanced infrastructure using Terraform for automated deployment of machine learning environments, ensuring consistency and reliability in model training and deployment.
- Led the development of a GraphQL-based API for efficient querying of pet product data, resulting in a 25% improvement in data retrieval speed for user-facing applications.

Jr Machine Learning Engineer-QUARTIC AI, (San Jose, California), Remote

Jul 2021 - Jun 2022

- Developed and deployed scalable ML models for predictive maintenance and operational optimization, utilizing Python, TensorFlow, and scikit-learn.
- Implemented algorithms for large-scale data analysis and production scheduling, reducing downtime by 25% and improving system efficiency.
- Led the design and implementation of Python-driven front-end solutions using **React** and **AngularJS** for real-time **risk monitoring**, improving UI responsiveness and accelerating decision-making for risk analysts by **20%**.

Artificial Intelligence Intern - HOPS Healthcare, India

Jun 2020 - Jun 2021

- Attained **85%** accuracy in mobile-based skin disease detection by implementing **TensorFlow** and **convolutional neural networks**, effectively enhancing diagnostic precision and contributing to advancements in healthcare technology.
- Collaborated with a team of data scientists to develop noise reduction algorithms using **Python** and **scikit-learn** for heart condition classification, enhancing diagnostic accuracy by **10%**.
- Improved COVID-19 lung classification accuracy by **20%** by implementing and ensembling machine learning and deep learning models in **Python**, utilizing **CNNs** for COVID/Not-COVID **X-ray analysis**.

EDUCATION:

Arizona State University Tempe, AZ

Aug 2022 - Dec 2023

Master's in Computer Science

GPA - 3.55

Pandit Deendayal Energy University, India

Jul 2018 - May 2022

Bachelor's in Information and Communication Technology

GPA - 3.75

PROJECTS:

Brain Tumor Recognizer App 🧠

Microsoft Azure Hackathon

- Developed a real-time AI model for 2D brain MRI classification using TensorFlow Lite and Azure's Custom Vision Service.
- Implemented Azure Machine Learning for automated model training and deployment, boosting efficiency by 20%.
- Integrated the TensorFlow Lite model into an Android app for mobile healthcare via smartphone camera.

Skin Disease Detector App 🧴

Google DSC Hackathon

- Led development of a skin disease detection app during a 24-hour hackathon, utilizing TensorFlow Lite for real-time classification.
- Preprocessed and augmented Harvard skin disease images with OpenCV, achieving 82% accuracy through model tuning.
- Integrated the model into an Android app for disease classification and finding nearby dermatologists.